# 2-Morita Equivalent Condensable Algebras in Topological Orders

Rongge Xu

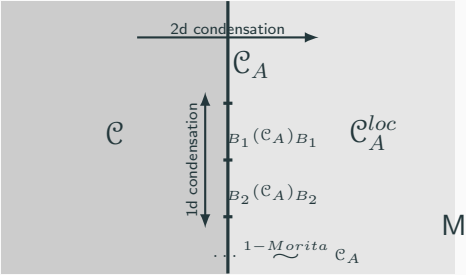
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Based on [arXiv: 2403.19779], a joint work with Holiverse Yang

## Primer [Kitaev06, Bais-Slingerland-Haaker09, Davydov10, Kitaev-Kong12, Davydov-Müger-Nikshych-Ostrik13, Kong14]

- A 2d anomaly-free topological order can be described by a unitary modular tensor category (MTC) $\mathcal{C}$ (with a central charge $c$).
- A 2d condensable (separable commutative Frobenius) algebra $A \in \mathcal{C}$ can be condensed to create new topological order, described by the category $\mathcal{C}_A^{loc}$.
- Anyons that are confined from going to the condensed phase form a 1d domain wall described by a fusion category $\mathcal{C}_A$.
- A 1d condensable algebra $B \in \mathcal{C}_A$ can be condensed to create another domain wall ${}_B(\mathcal{C}_A)_B$.




$$\mathcal{C}_{A_1}^{loc} \simeq \mathcal{C}_{A_2}^{loc}$$
$$\Updownarrow$$
$$A_1 \overset{2-Morita}{\sim} A_2$$

Motivation: classify 2-Morita equivalent $A_i$

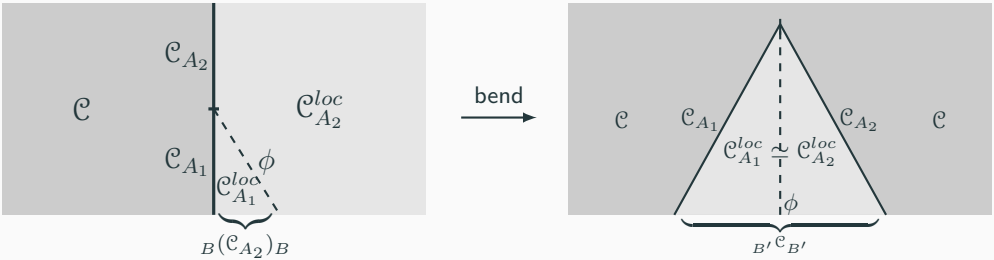
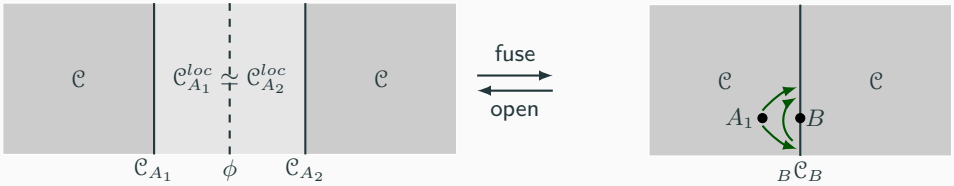



**Figure 1:** An interchange of anyons between $\mathcal{C}^{loc}_{A_2}$ and $\mathcal{C}^{loc}_{A_1}$ can be regarded as an invertible domain wall $\phi : \mathcal{C}^{loc}_{A_1} \xrightarrow{\sim} \mathcal{C}^{loc}_{A_2}$.

A gapped domain wall (codimension 1 defect) within a 2d topological order $\mathcal{C}$ can be described by $\mathcal{C}_{A_1} \boxtimes_{\mathcal{C}^{loc}_{A_1}} \phi \boxtimes_{\mathcal{C}^{loc}_{A_2}} \mathcal{C}_{A_2} \simeq {}_{B'}\mathcal{C}_{B'}$. $\phi : \mathcal{C}^{loc}_{A_1} \xrightarrow{\sim} \mathcal{C}^{loc}_{A_2}$ is characterized by an internal symmetry of $\mathcal{C}^{loc}_{A_1}$ (or $\mathcal{C}^{loc}_{A_2}$ equivalently), i.e. the group $\mathcal{A}\mathrm{ut}^{\mathrm{br}}_{\otimes}(\mathcal{C}^{loc}_{A_1})$ of braided auto-equivalence.

Classification of equivalent condensed phases can be translated to the classification of simple objects in 1 co-dimensional category of topological order.

Result I:




$$A_1 \simeq Z_l(B) \overset{2-Morita}{\underset{\phi}{\sim}} A_2 \simeq Z_r(B)$$

**Theorem (Based on [Fröhlich-Fuchs-Runkel-Schweigert 06])**

*The triples $(A_1, A_2, \phi)$ in $\mathcal{C}$ are classified by (1-Morita classes of) indecomposable separable algebras $B_i$ in $\mathcal{C}$, in which $A_1 \cong Z_l(B_i)$ and $A_2 \cong Z_r(B_i)$ represent the left and right centers of $B_i$.*

**Theorem (Based on [Davydov-Nikshych-Ostrik 12])**

*We have ${}_B\mathcal{C}_B \simeq \mathcal{C}_{Z_l(B)} \boxtimes_{\mathcal{C}^{loc}_{Z_l(B)}} \phi \boxtimes_{\mathcal{C}^{loc}_{Z_r(B)}} \mathcal{C}_{Z_r(B)}$ where $\phi : \mathcal{C}^{loc}_{Z_l(B)} \xrightarrow{\sim} \mathcal{C}^{loc}_{Z_r(B)}$ is an equivalence of MTCs.*

**Theorem**

$\mathcal{C}_A \boxtimes_{\mathcal{C}_A^{loc}} (\mathcal{C}_A^{loc})_{A'} \simeq \mathcal{C}_{A'}$ *as monoidal* $\mathcal{C}$-$\mathcal{C}_{A'}^{loc}$*-bimodule.*

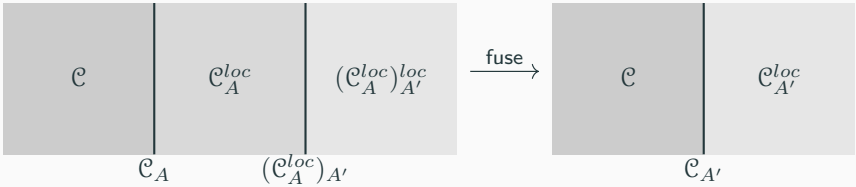
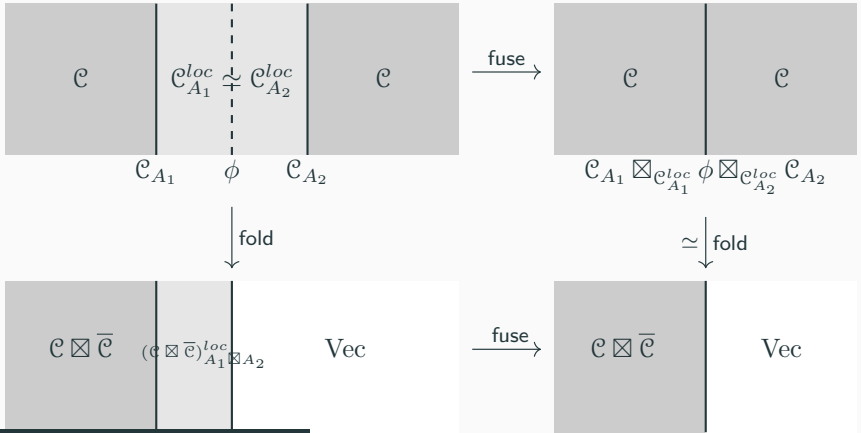



**Figure 2:** For two 2d condensable algebras $A$ and $A'$ with $A \hookrightarrow A'$ in $\mathcal{C}$, first condensing $A$ in $\mathcal{C}$ then $A'$ condensing $A'$ in $\mathcal{C}_A^{loc}$ gives the same phase as condensing $A'$ in $\mathcal{C}$ directly, i.e. $(\mathcal{C}_A^{loc})_{A'}^{loc} \simeq \mathcal{C}_{A'}^{loc}$. However, whether the gapped domain walls generated in the two step condensation after fusion (i.e. $\mathcal{C}_A \boxtimes_{\mathcal{C}_A^{loc}} (\mathcal{C}_A^{loc})_{A'}$) is equivalent to $\mathcal{C}_{A'}$ or not has not been discussed before.

**Lemma**

*Let $\mathcal{C}$ be a MTC. Given any pair of $E_2$-Morita equivalent algebras $A_1 \overset{2-Morita}{\sim} \phi A_2$, $\mathcal{C}_{A_1} \boxtimes_{\mathcal{C}^{loc}_{A_1}} \phi \boxtimes_{\mathcal{C}^{loc}_{A_2}} \mathcal{C}_{A_2}$ is equivalent to $\mathfrak{Z}(\mathcal{C})_L$ as monoidal $\mathcal{C}$-$\mathcal{C}$-bimodule for some lagrangian algebra $L \in \mathfrak{Z}(\mathcal{C})$.*



Example: 2d Toric code model $\mathcal{TC} := \mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_2))$ $\mathsf{H} := \sum_v (1 - \mathsf{A}_v) + \sum_p (1 - \mathsf{B}_p)$

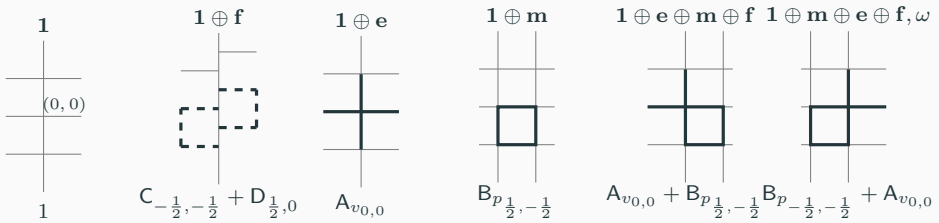


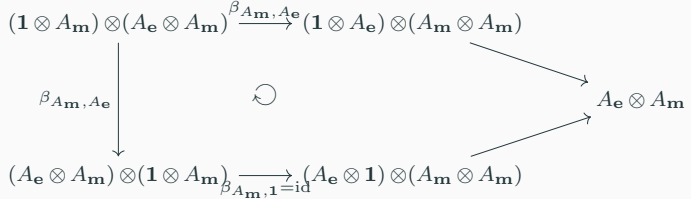

**Figure 4:** Lattice realizations of 1d condensable algebra $B_i$ locally. Condensing $B_i$ on trivial wall $\mathcal{TC}$ is equivalent to removing these thick edges for all $k$ along the neighborhood of column $0$.

For example, we can add a local trap $\mathsf{A}_{v_{0,0}} + \mathsf{B}_{p_{\frac{1}{2},-\frac{1}{2}}}$ to the original Hamiltonian H, and obtain

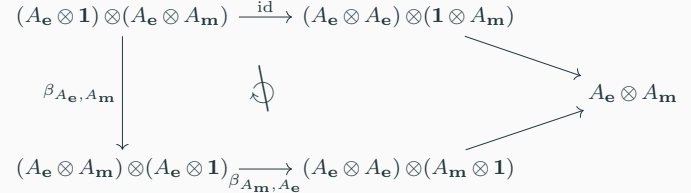
$$\mathsf{H}' := \mathsf{H} + \mathsf{A}_{v_{0,0}} + \mathsf{B}_{p_{\frac{1}{2},-\frac{1}{2}}} = \sum_{v \neq v_{0,0}} (1 - \mathsf{A}_v) + \sum_{p \neq p_{\frac{1}{2},-\frac{1}{2}}} (1 - \mathsf{B}_p) + 2$$

the topological excitation generated by the local trap $\mathsf{A}_{v_{0,0}} + \mathsf{B}_{p_{\frac{1}{2},-\frac{1}{2}}}$ is $\mathbf{1} \oplus \mathbf{e} \oplus \mathbf{m} \oplus \mathbf{f}$.

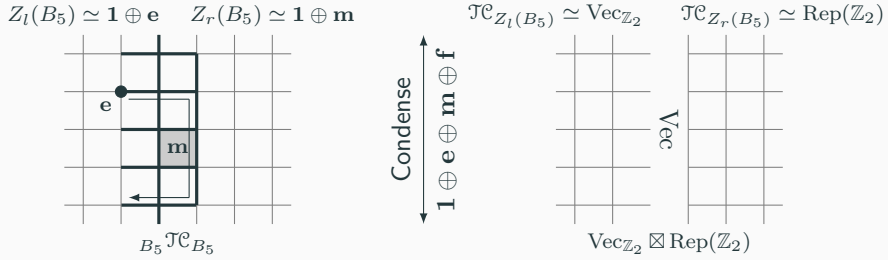
$B_5 = \mathbf{1} \oplus \mathbf{e} \oplus \mathbf{m} \oplus \mathbf{f}$ can be regarded as the tensor of two commutative algebras $A_\mathbf{e} := \mathbf{1} \oplus \mathbf{e}$ and $A_\mathbf{m} := \mathbf{1} \oplus \mathbf{m}$, i.e. $B_5 \cong (\mathbf{1} \oplus \mathbf{e}) \otimes (\mathbf{1} \oplus \mathbf{m})$. So there are two candidates $A_\mathbf{e}$ and $A_\mathbf{m}$ to consider. Here we prove that the subalgebra $\mathbf{1} \oplus \mathbf{m} \cong \mathbf{1} \otimes A_\mathbf{m}$ of $B_5 \cong A_\mathbf{e} \otimes A_\mathbf{m}$ is the Davydov's left center $C_l(B_5)$, and $\mathbf{1} \oplus \mathbf{e} \cong A_\mathbf{e} \otimes \mathbf{1}$ is not. Indeed, the following diagram commutes:




Hence we have $Z_r(B_5) \cong C_l(B_5) \cong A_\mathbf{m} = \mathbf{1} \oplus \mathbf{m}$. In addition, the following diagram does not commute,




since $\beta_{A_\mathbf{e}, A_\mathbf{m}} = \mathrm{id} \oplus \mathrm{id} \oplus \mathrm{id} \oplus \beta_{\mathbf{e},\mathbf{m}} = \mathrm{id} \oplus \mathrm{id} \oplus \mathrm{id} \oplus -\mathrm{id}$ and $\beta_{A_\mathbf{m}, A_\mathbf{e}} = \mathrm{id}$. So $\mathbf{1} \oplus \mathbf{e} \cong A_\mathbf{e} \otimes \mathbf{1}$ is not Braided [illegible] we $Z_r(B_5) \cong C_l(B_5) = \mathbf{1} \oplus \mathbf{e}$. Therefore

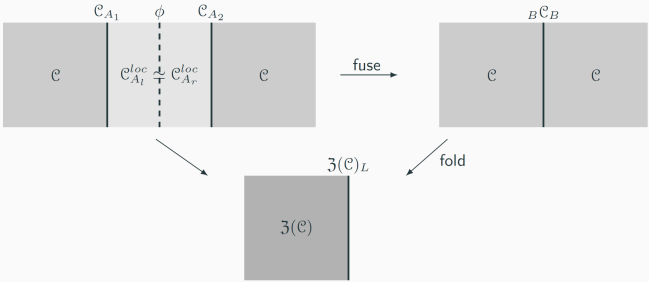



$$\mathsf{H}_{wall} = \mathsf{H} + \sum_k \mathsf{A}_{v_{0,k}} + \sum_k \mathsf{B}_{p_{\frac{1}{2},k-\frac{1}{2}}} = \sum_{v \neq v_{0,k}} (1 - \mathsf{A}_v) + \sum_{p \neq p_{\frac{1}{2},k-\frac{1}{2}}} (1 - \mathsf{B}_p) + 2N$$

$B_5$ located on the domain wall can not expand itself freely into the 2d bulk due to the non-trivial braiding $\beta_{\mathbf{e},\mathbf{m}} = -1$ in $\mathcal{TC}$. However, sub-algebra $\mathbf{1} \oplus \mathbf{e}$ can be expanded to the left bulk of the domain wall. The half braiding $\beta_{\mathbf{m},\mathbf{e}} = 1$ from the left side of the wall is trivial. $\mathbf{1} \oplus \mathbf{m}$ in this case is blocked by the structure of wall from going to the right bulk. This shows $Z_l(B_5)$ should be $\mathbf{1} \oplus \mathbf{e}$.

## Result III:

The pairs of 2-Morita equivalent condensable algebras in $\mathcal{C}$ can also be classified by lagrangian algebras (maximal 2d condensable algebras) $L_i$'s in $\mathcal{C} \boxtimes \overline{\mathcal{C}} \simeq \mathfrak{Z}(\mathcal{C})$.




$$L_i \cap (\mathcal{C} \boxtimes \mathbf{1}) =: A_{l_i} \overset{2-Morita}{\sim} A_{r_i} := L_i \cap (\mathbf{1} \boxtimes \overline{\mathcal{C}})$$

**Theorem**

*Given any pair of 2-Morita equivalent algebras* $(A_l, A_r)$ *in* $\mathcal{C}$, $\mathcal{C}_{A_l} \boxtimes_{\mathcal{C}^{loc}_{A_l}} \phi \boxtimes_{\mathcal{C}^{loc}_{A_r}} \mathcal{C}_{A_r}$ *is equivalent to* $\mathfrak{Z}(\mathcal{C})_L (\simeq \mathfrak{Z}(\mathcal{C})_{Z(B)}$ *[Kong-Runkel 09]) as monoidal* $\mathcal{C}$-$\mathcal{C}$*-bimodule for some lagrangian algebra* $L \in \mathfrak{Z}(\mathcal{C})$.

| $B_i \in \mathcal{TC}$ | $Z_l(B_i)/Z_r(B_i)$ | Domain wall | Lagrangian algebras $L_i \in \mathcal{TC} \boxtimes \overline{\mathcal{TC}}$ |
|---|---|---|---|
| $\mathbf{1}$ | $\mathbf{1}/\mathbf{1}$ | trivial wall | $\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{e}\bar{\mathbf{e}} \oplus \mathbf{m}\bar{\mathbf{m}} \oplus \mathbf{f}\bar{\mathbf{f}}$ |
| $\mathbf{1} \oplus \mathbf{f}$ | $\mathbf{1}/\mathbf{1}$ | $\mathbf{e} - \mathbf{m}$ exchange | $\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{m}\bar{\mathbf{e}} \oplus \mathbf{e}\bar{\mathbf{m}} \oplus \mathbf{f}\bar{\mathbf{f}}$ |
| $\mathbf{1} \oplus \mathbf{e}$ | $\mathbf{1} \oplus \mathbf{e}/\mathbf{1} \oplus \mathbf{e}$ | $\mathrm{Vec}_{\mathbb{Z}_2} \boxtimes \mathrm{Vec}_{\mathbb{Z}_2}$ | $\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{e}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{e}} \oplus \mathbf{e}\bar{\mathbf{e}}$ |
| $\mathbf{1} \oplus \mathbf{m}$ | $\mathbf{1} \oplus \mathbf{m}/\mathbf{1} \oplus \mathbf{m}$ | $\mathrm{Rep}(\mathbb{Z}_2) \boxtimes \mathrm{Rep}(\mathbb{Z}_2)$ | $\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{m}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{m}} \oplus \mathbf{m}\bar{\mathbf{m}}$ |
| $\mathbf{1} \oplus \mathbf{e} \oplus \mathbf{m} \oplus \mathbf{f}$ | $\mathbf{1} \oplus \mathbf{e}/\mathbf{1} \oplus \mathbf{m}$ | $\mathrm{Vec}_{\mathbb{Z}_2} \boxtimes \mathrm{Rep}(\mathbb{Z}_2)$ | $\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{e}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{m}} \oplus \mathbf{e}\bar{\mathbf{m}}$ |
| $\mathbf{1} \oplus \mathbf{e} \oplus \mathbf{m} \oplus \mathbf{f}, \omega$ | $\mathbf{1} \oplus \mathbf{m}/\mathbf{1} \oplus \mathbf{e}$ | $\mathrm{Rep}(\mathbb{Z}_2) \boxtimes \mathrm{Vec}_{\mathbb{Z}_2}$ | $\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{m}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{e}} \oplus \mathbf{m}\bar{\mathbf{e}}$ |

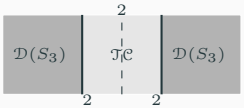
| $H$ | $F$ | $E_2$ condensable algebras in $\mathcal{TC}$ | Condensed phase $\mathcal{TC}_A^{loc}$ | Domain walls | Total: 6 |
|---|---|---|---|---|---|
| $\mathbb{Z}_2$<br>$\{e\}$ | $\mathbb{Z}_2$<br>$\{e\}$ | $\mathbf{1} \oplus \mathbf{m}$<br>$\mathbf{1} \oplus \mathbf{e}$ | Vec |  | non-invertible:<br>$2 \times 2 = 4$ |
| $\mathbb{Z}_2$ | $\{e\}$ | $\mathbf{1}$ | $\mathcal{TC}$ |  | invertible: 2 |

there are 22 gapped domain walls in $\mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_4)) := \mathcal{D}(\mathbb{Z}_4)$ in total.

| $H$ | $F$ | Condensable algebras in $\mathcal{D}(\mathbb{Z}_4)$ | Condensed phase $\mathcal{D}(\mathbb{Z}_4)^{loc}_A$ | Domain walls | Total: 22 |
|---|---|---|---|---|---|
| $\{e\}$<br>$\mathbb{Z}_2$<br>$\mathbb{Z}_4$ | $\{e\}$<br>$\mathbb{Z}_2$<br>$\mathbb{Z}_4$ | $\mathbf{1}\oplus\mathbf{e}\oplus\mathbf{e}^2\oplus\mathbf{e}^3$<br>$\mathbf{1}\oplus\mathbf{e}^2\oplus\mathbf{m}^2\oplus\mathbf{f}^2$<br>$\mathbf{1}\oplus\mathbf{m}\oplus\mathbf{m}^2\oplus\mathbf{m}^3$ | Vec | $\mathcal{D}(\mathbb{Z}_4)$ \| Vec \| $\mathcal{D}(\mathbb{Z}_4)$ (3, 3) | non-invertible: $3\times 3 = 9$ |
| $\mathbb{Z}_4$ | $\mathbb{Z}_2$ | $\mathbf{1}\oplus\mathbf{f}^2$ | $\mathcal{DS}$ | $\mathcal{D}(\mathbb{Z}_4)$ \| $\mathcal{DS}$ \| $\mathcal{D}(\mathbb{Z}_4)$ | non-invertible: 1 |
| $\mathbb{Z}_4$<br>$\mathbb{Z}_2$ | $\mathbb{Z}_2$<br>$\{e\}$ | $\mathbf{1}\oplus\mathbf{m}^2$<br>$\mathbf{1}\oplus\mathbf{e}^2$ | $\mathcal{TC}$ | $\mathcal{D}(\mathbb{Z}_4)$ \| $\mathcal{TC}$ \| $\mathcal{D}(\mathbb{Z}_4)$ (2, 2, 2) | non-invertible: $2\times 2\times 2 = 8$ |
| $\mathbb{Z}_4$ | $\{e\}$ | $\mathbf{1}$ | $\mathcal{D}(\mathbb{Z}_4)$ | $\mathcal{D}(\mathbb{Z}_4)$ \| $\mathcal{D}(\mathbb{Z}_4)$ (4) | invertible: 4 |

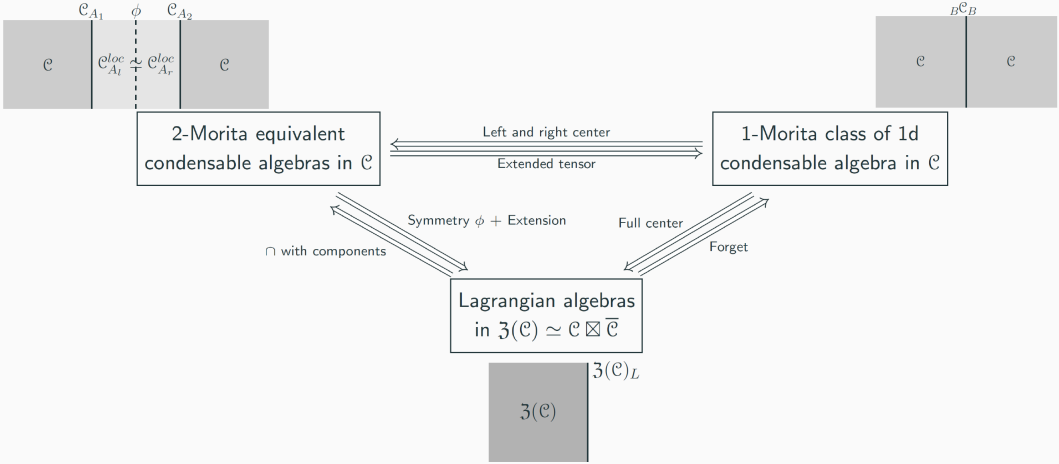
**Table 1:** Seven 2d condensable algebras in $\mathcal{D}(\mathbb{Z}_4)$ are listed in the third column of the table. There are four 2-Morita equivalence classes of condensable algebra that condense to Vec, $\mathcal{DS}$, $\mathcal{TC}$, $\mathcal{D}(\mathbb{Z}_4)$ respectively. We have 22 domain walls in total (drawn in the fifth column),

Results in $\mathfrak{Z}(\mathrm{Rep}(S_3)) := \mathcal{D}(S_3)$

| $H$ | $F$ | Condensable algebras in $\mathcal{D}(S_3)$ | Condensed phase $\mathcal{D}(S_3)^{loc}_{A}$ | Domain walls | Total: 28 |
|---|---|---|---|---|---|
| $S_3$<br>$A_3$<br>$C_2$<br>$\{e\}$ | $S_3$<br>$A_3$<br>$C_2$<br>$\{e\}$ | $\mathbf{A}\oplus\mathbf{F}\oplus\mathbf{D}$<br>$\mathbf{A}\oplus\mathbf{B}\oplus 2\mathbf{F}$<br>$\mathbf{A}\oplus\mathbf{C}\oplus\mathbf{D}$<br>$\mathbf{A}\oplus\mathbf{B}\oplus 2\mathbf{C}$ | Vec |  | non-invertible: $4\times 4 = 16$ |
| $S_3$<br>$C_2$ | $A_3$<br>$\{e\}$ | $\mathbf{A}\oplus\mathbf{F}$<br>$\mathbf{A}\oplus\mathbf{C}$ | $\mathcal{TC}$ |  | non-invertible: $2\times 2\times 2 = 8$ |
| $A_3$ | $\{e\}$ | $\mathbf{A}\oplus\mathbf{B}$ | $\mathcal{D}(\mathbb{Z}_3)$ |  | non-invertible: $4/2 = 2$ |
| $S_3$ | $\{e\}$ | $\mathbf{A}$ | $\mathcal{D}(S_3)$ |  | invertible: 2 |

## Main results:

We give a complete interplay between $E_1$ condensable algebras in $\mathcal{C}$, 2-Morita equivalent $E_2$ condensable algebras in $\mathcal{C}$, and lagrangian algebras in $\mathcal{C} \boxtimes \overline{\mathcal{C}}$.

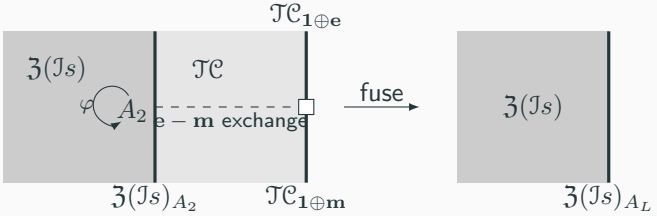


# Double Ising

Algebra automorphism

Consider the Ising topological order $\mathcal{I}s$ with anyon $\mathbf{1}$, $\psi$, $\sigma$ and the fusion rules are given by: $\sigma\otimes\sigma=\mathbf{1}\oplus\psi, \sigma\otimes\psi=\sigma, \psi\otimes\psi=\mathbf{1}$. Double Ising $\mathfrak{Z}(\mathcal{I}s)$ admit three 2d condensable algebras $A_1=\mathbf{1}\boxtimes\overline{\mathbf{1}}$, $A_2=(\mathbf{1}\boxtimes\overline{\mathbf{1}})\oplus(\psi\boxtimes\overline{\psi})$ and $A_L=(\mathbf{1}\boxtimes\overline{\mathbf{1}})\oplus(\sigma\boxtimes\overline{\sigma})\oplus(\psi\boxtimes\overline{\psi})$, which trivially condense to $\mathfrak{Z}(\mathcal{I}s)$ itself, $\mathcal{TC}$ and $\mathrm{Vec}$ respectively [**?**].

Double Ising has three inequivalent gapped domain walls: $\mathfrak{Z}(\mathcal{I}s)\simeq\mathcal{I}s\boxtimes\overline{\mathcal{I}s}$ is the trivial domain wall, and $\mathcal{I}s\boxtimes\mathcal{I}s$ is the wall induced by condensing lagrangian algebra $A_L$. Although $\mathcal{TC}$ has an $\mathbf{e}-\mathbf{m}$ exchange domain wall, $\mathfrak{Z}(\mathcal{I}s)_{A_2}\boxtimes_{\mathcal{TC}}\mathfrak{Z}(\mathcal{I}s)_{A_2}$ is the unique domain wall associated to the 2-Morita class $(\mathbf{1}\boxtimes\overline{\mathbf{1}})\oplus(\psi\boxtimes\overline{\psi})$. This is due to the $\mathbf{e}$-$\mathbf{m}$ exchange symmetry in $\mathfrak{Z}(\mathcal{I}s)^{loc}_{A_2}\simeq\mathcal{TC}$ is induced by the non-trivial algebra automorphism of $A_2$.



$$\mathcal{D}(S_3)^{loc}_{\mathbf{A}\oplus\mathbf{B}} \xrightarrow{\sim} \mathcal{D}(\mathbb{Z}_3)$$

$$\mathbf{A}\oplus\mathbf{B} \mapsto \mathbf{1}$$

$$\mathbf{C} \mapsto \mathbf{e} \qquad \mathbf{C}^{tw} \mapsto \mathbf{e}^2$$

$$\mathbf{F} \mapsto \mathbf{m} \qquad \mathbf{F}^{tw} \mapsto \mathbf{m}^2$$

$$\mathbf{G} \mapsto \mathbf{em} \qquad \mathbf{G}^{tw} \mapsto \mathbf{e}^2\mathbf{m}^2$$

$$\mathbf{H} \mapsto \mathbf{e}^2\mathbf{m} \qquad \mathbf{H}^{tw} \mapsto \mathbf{em}^2$$

